

Topic 13

Excretion in humans

Concept guide · Worked application · Cambridge-style practice · Fully explained answers

What this guide teaches

A complete explanation of the syllabus concepts, biological mechanisms, common misconceptions, data interpretation and original exam-style practice for this topic.

Syllabus map

13.1	Excretion
13.2	Urinary system
13.3	Nephron function
13.4	Dialysis and transplantation

How to study: retrieve definitions, cover the application, then complete the practice paper without using the notes.

1. Excretion

Excretion removes toxic materials, waste products of metabolism and substances in excess. Carbon dioxide is excreted by lungs; urea is formed in the liver from excess amino acids and excreted by kidneys; excess water and ions leave through kidneys and skin. Egestion is removal of undigested food and is not excretion.

Examiner insight

Name the metabolic origin of a waste when possible.

2. Kidney and nephron

Renal arteries deliver blood; renal veins remove filtered blood; ureters carry urine to the bladder; the urethra releases urine. Each nephron contains a glomerulus, Bowman's capsule, tubule and collecting duct.

Examiner insight

Do not confuse ureter, kidney to bladder, with urethra, bladder to outside.

3. Filtration and reabsorption

High pressure in the glomerulus filters water, urea, ions and glucose into Bowman's capsule; cells and large proteins remain in blood. Selective reabsorption returns all glucose, needed ions and much water to blood. Urea remains concentrated in urine.

Examiner insight

Ultrafiltration is driven by pressure; selective reabsorption uses specific transport processes.

4. Kidney failure

Dialysis uses a partially permeable membrane and dialysis fluid with normal glucose and ion concentrations but no urea, creating gradients that remove waste without losing useful solutes. Transplants offer greater freedom but require surgery, a donor and immunosuppressants.

Examiner insight

Compare treatments using frequency, cost, rejection risk and quality of life.

Worked application

Situation: A patient produces urine containing glucose.

Prediction: Selective reabsorption capacity may be exceeded or impaired.

Reason: Normally all filtered glucose is reabsorbed into blood.

Exam technique

High-value habits

Define terms precisely, describe visible patterns before explaining them, link structure to function, use data values as evidence, identify variables explicitly and answer the command word directly.

Retrieval checklist

- Recall five key definitions without notes.
- Explain one process as a linked cause-and-effect chain.
- Interpret the data set and identify a limitation.
- Complete the timed practice and mark only after finishing.

Practice Paper

Topic 13 · Excretion in humans

Instructions

Answer all questions. Select one option for multiple choice. Use values from tables where relevant. Write explanations on the lines. Individual parts carry 1-3 marks.

Candidate name:

Class / date:

Total marks: 50

Suggested time: 60 minutes.

Section A · Multiple choice

Choose the best answer for each question.

1. Which statement best defines excretion? [1]
- A removal of metabolic waste, toxic material and excess substances
 - B a nitrogenous waste made in the liver from excess amino acids
 - C pressure filtration of small molecules from glomerular blood
 - D return of useful substances from nephron filtrate to blood
2. Which statement best defines urea? [1]
- A pressure filtration of small molecules from glomerular blood
 - B a nitrogenous waste made in the liver from excess amino acids
 - C return of useful substances from nephron filtrate to blood
 - D artificial removal of wastes and excess ions or water across a membrane
3. Which statement best defines ultrafiltration? [1]
- A return of useful substances from nephron filtrate to blood
 - B artificial removal of wastes and excess ions or water across a membrane
 - C pressure filtration of small molecules from glomerular blood
 - D removal of metabolic waste, toxic material and excess substances
4. Which statement best defines selective reabsorption? [1]
- A artificial removal of wastes and excess ions or water across a membrane
 - B removal of metabolic waste, toxic material and excess substances
 - C a nitrogenous waste made in the liver from excess amino acids
 - D return of useful substances from nephron filtrate to blood
5. Which statement best defines dialysis? [1]
- A artificial removal of wastes and excess ions or water across a membrane
 - B removal of metabolic waste, toxic material and excess substances
 - C a nitrogenous waste made in the liver from excess amino acids
 - D pressure filtration of small molecules from glomerular blood
6. Which statement about excretion is correct? [1]
- A Removal of undigested food.
 - B Excretion removes toxic materials, waste products of metabolism and substances in excess.
 - C A digestive enzyme made in kidneys.
 - D Active transport of glucose into urine.

7. Which statement about kidney and nephron is correct? [1]
- A A digestive enzyme made in kidneys.
 - B Active transport of glucose into urine.
 - C Renal arteries deliver blood; renal veins remove filtered blood; ureters carry urine to the bladder; the urethra releases urine.
 - D Filtration of all blood cells.
8. Which statement about filtration and reabsorption is correct? [1]
- A Active transport of glucose into urine.
 - B Filtration of all blood cells.
 - C Surgical replacement of a kidney.
 - D High pressure in the glomerulus filters water, urea, ions and glucose into Bowman's capsule; cells and large proteins remain in blood.
9. Which statement about kidney failure is correct? [1]
- A Dialysis uses a partially permeable membrane and dialysis fluid with normal glucose and ion concentrations but no urea, creating gradients that remove waste without losing useful solutes.
 - B Filtration of all blood cells.
 - C Surgical replacement of a kidney.
 - D Removal of undigested food.
10. Which prediction is most appropriate when a patient produces urine containing glucose? [1]
- A Removal of undigested food.
 - B Active transport of glucose into urine.
 - C Selective reabsorption capacity may be exceeded or impaired.
 - D Surgical replacement of a kidney.

Section B · Structured questions

11 Biologists use precise vocabulary when studying excretion in humans.

(a) Define excretion.

[2]

(b) Define urea.

[2]

(c) Distinguish ultrafiltration from the incorrect idea: “active transport of glucose into urine”.

[2]

(d) State two important features of excretion.

[2]

[Total: 8]

12 A patient produces urine containing glucose.

(a) State the most likely biological outcome.

[2]

(b) Explain the outcome using a linked biological mechanism.

[3]

(c) Predict what would happen if the relevant change were reversed. [2]

(d) State one assumption made in this prediction. [1]

[Total: 8]

13 A student records relative urea concentration under three conditions.

condition	relative urea concentration
renal artery	6
renal vein	4
urine	55

(a) Calculate the difference between the highest and lowest recorded values. [2]

(b) Describe the pattern in the results. [2]

(c) Explain the pattern using biological knowledge. [3]

(d) State one limitation of drawing a conclusion from these data. [1]

[Total: 8]

14 A student plans to model dialysis using Visking tubing containing urea and glucose.

(a) Identify the independent variable. [1]

(b) Identify the dependent variable. [1]

(c) State two control variables. [2]

(d) Describe how to obtain reliable results across a suitable range. [3]

(e) Suggest one specific improvement. [1]

[Total: 8]

15 Apply your knowledge of excretion in humans to an unfamiliar context.

(a) Define selective reabsorption. [2]

(b) Explain the biological importance of filtration and reabsorption. [3]

(c) Correct this statement: “surgical replacement of a kidney”.

[2]

(d) State one further question that could be investigated.

[1]

[Total: 8]

Worked solutions

Remove or fold back this section while completing the practice paper.

Marking routine

Mark each part separately. Award points for accurate biology, use of evidence and linked reasoning, then diagnose the first error.

Solutions · Multiple choice

- 1 A. Removal of metabolic waste, toxic material and excess substances.
- 2 B. A nitrogenous waste made in the liver from excess amino acids.
- 3 C. Pressure filtration of small molecules from glomerular blood.
- 4 D. Return of useful substances from nephron filtrate to blood.
- 5 A. Artificial removal of wastes and excess ions or water across a membrane.
- 6 B. Name the metabolic origin of a waste when possible.
- 7 C. Do not confuse ureter, kidney to bladder, with urethra, bladder to outside.
- 8 D. Ultrafiltration is driven by pressure; selective reabsorption uses specific transport processes.
- 9 A. Compare treatments using frequency, cost, rejection risk and quality of life.
- 10 C. Normally all filtered glucose is reabsorbed into blood.

Solutions · Structured questions

11. (a) removal of metabolic waste, toxic material and excess substances [2]. (b) a nitrogenous waste made in the liver from excess amino acids [2]. (c) pressure filtration of small molecules from glomerular blood; not active transport of glucose into urine [2]. (d) Credit any two accurate features from the guide: Excretion removes toxic materials, waste products of metabolism and substances in excess. Carbon dioxide is excreted by lungs; urea is formed in the liver from excess amino acids and excreted by kidneys; excess water and ions leave through kidneys and skin. Egestion is removal of undigested food and is not excretion. [2].

12. (a) Selective reabsorption capacity may be exceeded or impaired [2]. (b) Normally all filtered glucose is reabsorbed into blood [3]. (c) Accept the biologically justified opposite trend when the causal factor is reversed [2]. (d) One valid assumption, such as other relevant conditions remaining constant [1].

13. (a) $55 - 4 = 51$ [2]. (b) The values change from 6 to 55, with the highest value 55. [2]. (c) Kidneys remove urea from blood and water reabsorption concentrates it in urine [3]. (d) Only three conditions / no repeats / small dataset / uncontrolled variation [1].

14. (a) External fluid composition or time [1]. (b) Urea concentration outside tubing [1]. (c) Same membrane area, volumes, temperature and starting concentrations [2]. (d) Use several values across a sensible range, repeat each condition, identify anomalies and calculate means [3]. (e) Stir solutions and use quantitative reagent tests [1].

15. (a) return of useful substances from nephron filtrate to blood [2]. (b) Award linked points from: High pressure in the glomerulus filters water, urea, ions and glucose into Bowman's capsule; cells and large proteins remain in blood. Selective reabsorption returns all glucose, needed ions and much water to blood. Urea remains concentrated in urine. [3]. (c) Correct statement: dialysis means artificial removal of wastes and excess ions or water across a membrane [2]. (d) Any testable relevant biological question [1].

Self-assessment

Classify the first lost mark as definition, biological knowledge, causal link, data use, experimental design or communication.

Evidence used: Cambridge IGCSE Biology 0610 syllabus 2026-2028 and June 2026 papers, mark schemes and Principal Examiner Report supplied by the user. All practice questions are newly written.